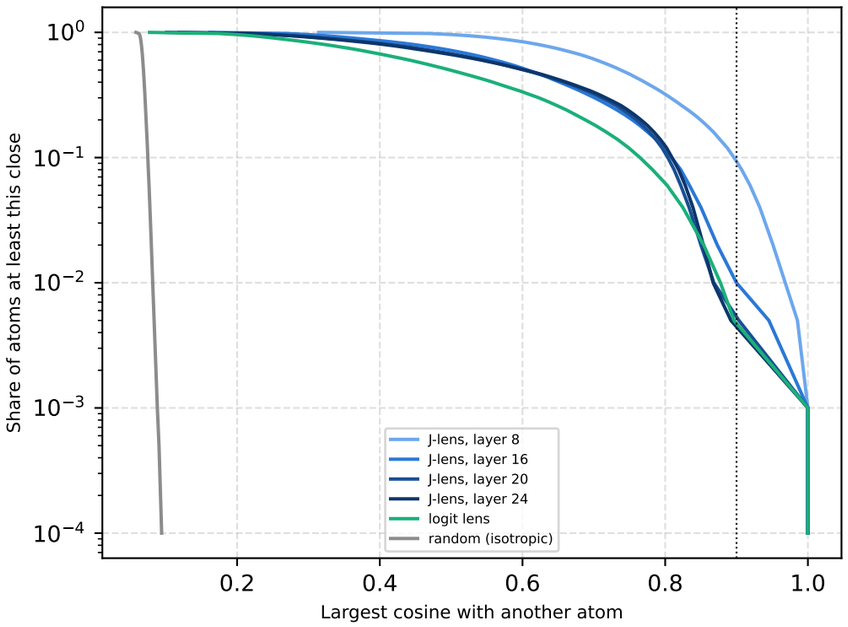
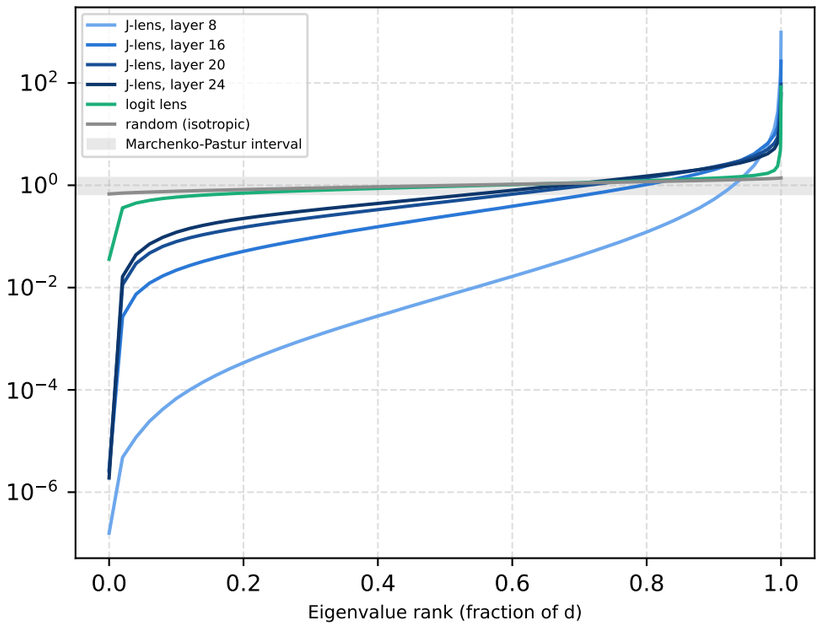
llama3.1-8b: N = 128,256 atoms in d = 4096

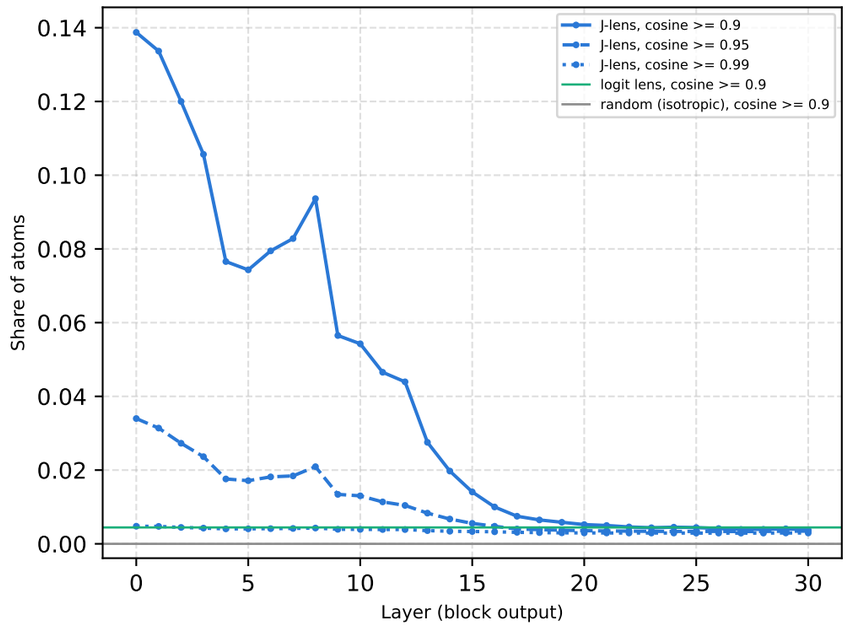
Nearest neighbour of each atom



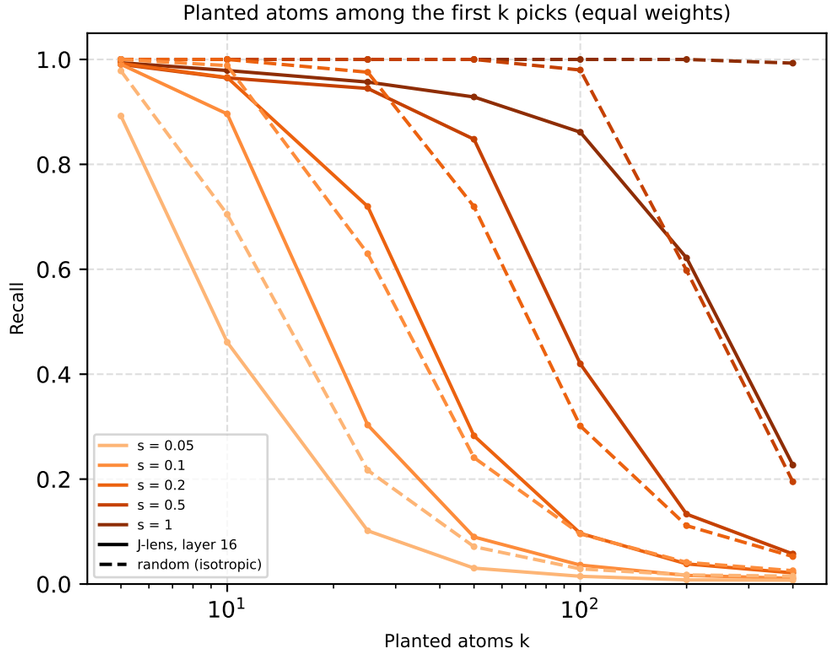
Spectrum of (d / N) $V^T V$ (mean 1)



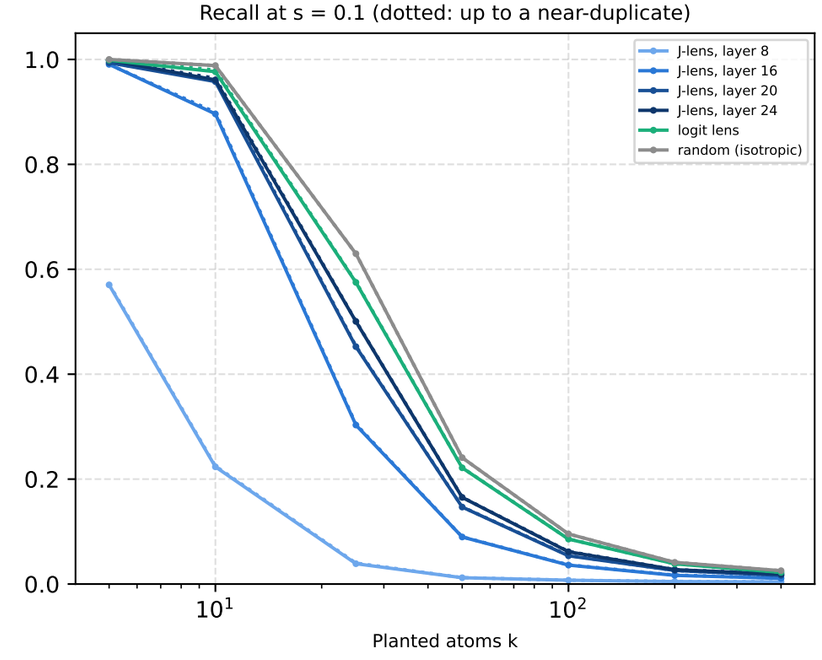
Atoms with a near-duplicate



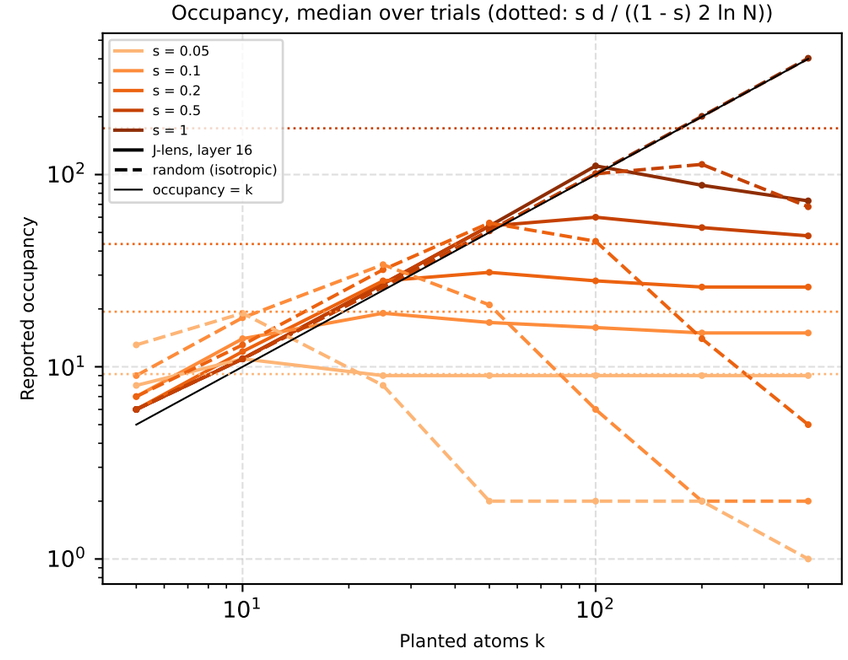
Planted atoms among the first k picks (equal weights)



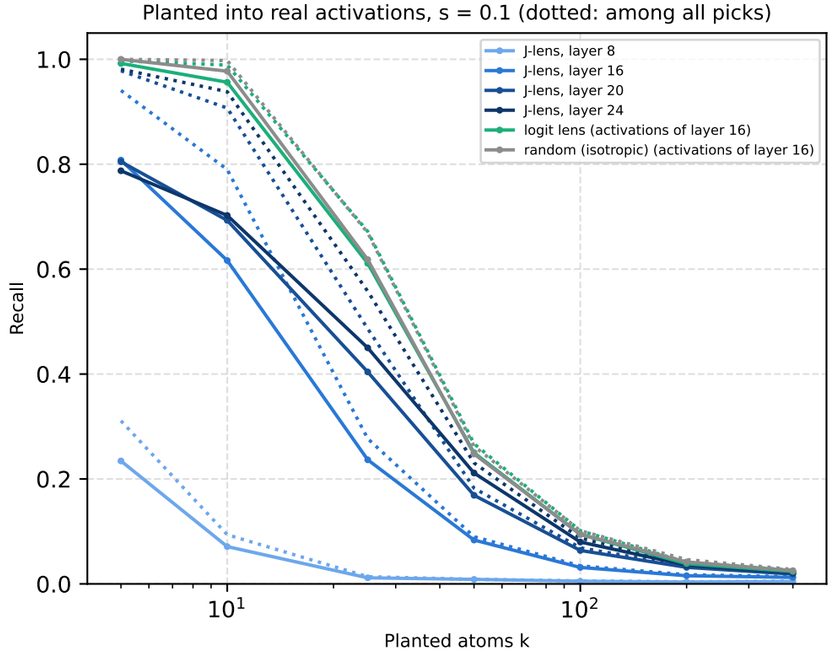
Recall at s = 0.1 (dotted: up to a near-duplicate)



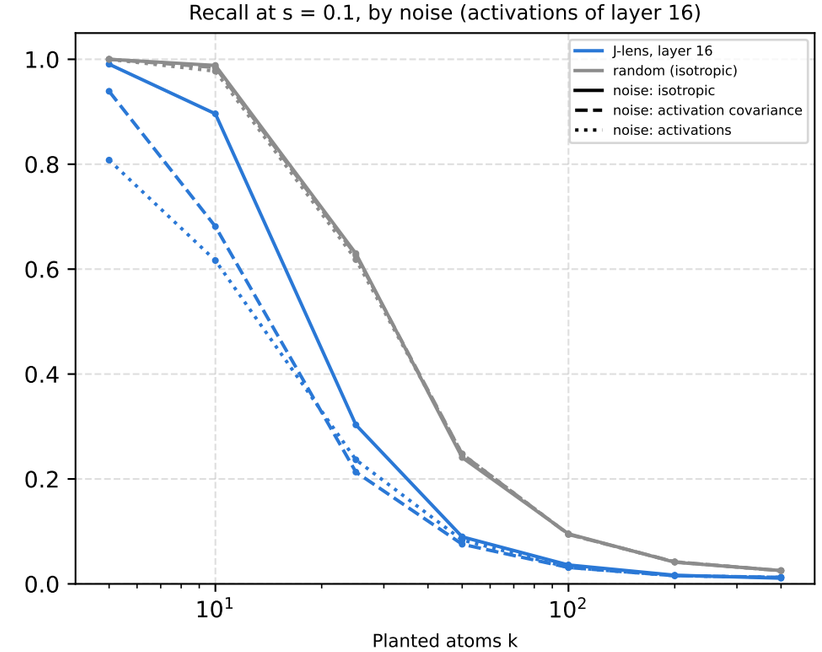
Occupancy, median over trials (dotted: s d / ((1 - s) 2 ln N))



Planted into real activations, s = 0.1 (dotted: among all picks)



Recall at s = 0.1, by noise (activations of layer 16)



Occupancy, median over trials, J-lens layer 16

